

Foysal Mahamud

BSc Computer Science & Engineering | North South University

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PROFILE

CSE undergraduate at NSU with proven ability to ship production-grade software. Built and deployed a multi-role Django + React grant platform used by a real university committee. Strong in full-stack web, C++ systems, and OOP.

TECHNICAL SKILLS

Languages

- Python
- Java
- C / C++
- TypeScript
- JavaScript
- SQL

Frontend

- React 19
- HTML5 / CSS3
- TailwindCSS
- Vite

Backend

- Django 4.2
- REST Framework
- JWT / Celery
- Redis

Databases

- PostgreSQL
- MySQL
- Supabase

Tools

- Git & GitHub
- Docker (basics)
- Railway
- Vercel

Other

- STM32 Embedded C
- OOP & DSA
- Agile / Scrum

EDUCATION

B.Sc. CS & Engineering
North South University
2023–Present • CGPA: 3.10/4.00 (84%)

Higher Secondary (HSC)
Science Group
2020 • GPA: 5.00/5.00

Secondary (SSC)
2018 • GPA: 5.00/5.00

SOFT SKILLS

- Problem Solving
- Team Collaboration
- Technical Communication
- Attention to Detail

PROFESSIONAL SUMMARY

Full-stack developer and CSE undergraduate with a record of shipping production-level software. Built the **CTRG Grant Review Management System** — a Django + React platform with FSM-driven proposal workflows, JWT auth, Celery background tasks, and auto-generated PDF/DOCX reports — deployed on Railway and Vercel. Also experienced in **C++ systems programming**, Java OOP, MySQL, and STM32 embedded systems. Writes clean, maintainable code and owns the full development lifecycle.

PROJECTS

CTRG Grant Review Management System

Django 4.2 · React 19 · TypeScript · PostgreSQL · Celery · JWT CSE 299 — Junior Design
■ github.com/foysal-mahamud/CSE-299-Junior-Design-Course-CTRG-management

- Designed a multi-role platform (PI / Reviewer / SRC Chair) managing the full grant proposal lifecycle: draft → submit → Stage 1 review → revision → Stage 2 review → final decision.
- Implemented Django FSM state machine with django-fsm-log for enforced, auditable state transitions; all business logic strictly server-side.
- Integrated Celery + Redis for automated deadline reminders; ReportLab & python-docx for downloadable PDF/DOCX review reports.
- Deployed React SPA on Vercel, Django API on Railway, PostgreSQL + file storage on Supabase — zero-downtime production environment.

Olympic Athlete Tracking System

C++17 · CMake · CSV I/O · Chart.js · GitHub Pages CSE 225 — Data Structures
■ github.com/foysal-mahamud/CSE225OlympicProject

- CLI application managing Paris 2024 Olympic records in a custom singly-linked list with full CRUD, pagination, and dual search modes.
- Wrote an RFC 4180-compliant CSV tokeniser for 30-column quoted data; discovered and fixed 6 column-mapping bugs in original code.
- Built companion web dashboard (Chart.js) deployed to GitHub Pages showing medal breakdowns, top athletes, and sport statistics.

Stock Buy & Sell Desktop Application

Java · MySQL · JDBC · Swing CSE 311 — Database Systems

- Relational schema for stock trading with user auth, live portfolio tracking, and transaction history; JDBC with parameterised queries to prevent SQL injection.

Car Racing Game

Java · OOP · Event Handling CSE 215 — Prog. Language II

- Interactive racing game with OOP design patterns, real-time keyboard events, collision detection, and progressive difficulty scaling.

7-Segment Display Controller

STM32 · Embedded C · Combinational Logic CSE 231 — Digital Logic Design

- Designed combinational + sequential circuits to drive a 7-segment display via STM32 MCU; programmed in Embedded C with direct register manipulation.

RELEVANT COURSEWORK

CSE 225 — Data Structures & Algorithms
Linked lists, trees, sorting, hashing, complexity analysis

CSE 311 — Database Systems
ER modelling, normalisation, SQL, JDBC, transactions

CSE 327 — Software Engineering
SDLC, Agile/Scrum, sprint planning, Kanban, XP

CSE 231 — Digital Logic Design
Boolean algebra, combinational/sequential circuits, STM32 lab

CSE 215 — Programming Language II
Advanced Java OOP, generics, collections, GUI programming